GROUP 03: Effect of Image Scaling and Zoom Augmentation

WEEK 2

OBJECTIVE FOR WEEK 2: Design Low/Medium/High intensity levels for the assigned Scaling/Zoom augmentation with justification; implement the augmentation in code; generate and visually inspect before/after sample grids for each intensity level.

WORK COMPLETED:

- Studied how scaling/zoom augmentation should work (resize about center, crop for zoom-in, pad for zoom-out) and worked out the bounding box math for it.
- Implemented the augmentation in Python/OpenCV, including trimming/removing boxes that fall outside the frame after zooming in.
- Chose three intensity levels based on realistic vehicle size variation:
    - Low: scale 0.90–1.10
    - Medium: scale 0.75–1.25
    - High: scale 0.60–1.40
- Generated before/after sample grids for each intensity and checked that bounding boxes tracked the vehicles correctly.
- Saved the sample outputs for the weekly report.

RESULTS:

1. Augmentation Implementation Code

```
def scale_zoom_augment(img, boxes, scale, pad_color=(114, 114, 114)):

    h, w = img.shape[:2]
    new_w, new_h = max(1, int(round(w * scale))), max(1, int(round(h * scale)))
    resized = cv2.resize(img, (new_w, new_h), interpolation=cv2.INTER_LINEAR)

    if scale >= 1:
        x0 = (new_w - w) // 2
        y0 = (new_h - h) // 2
        out_img = resized[y0:y0 + h, x0:x0 + w]
    else:
        out_img = np.full((h, w, 3), pad_color, dtype=np.uint8)
        x0 = (w - new_w) // 2
        y0 = (h - new_h) // 2
```

```
        out_img[y0:y0 + new_h, x0:x0 + new_w] = resized

    new_boxes = []
    for cls, xc, yc, bw, bh in boxes:
        xc2 = 0.5 + (xc - 0.5) * scale
        yc2 = 0.5 + (yc - 0.5) * scale
        bw2 = bw * scale
        bh2 = bh * scale

        x1, y1 = xc2 - bw2 / 2, yc2 - bh2 / 2
        x2, y2 = xc2 + bw2 / 2, yc2 + bh2 / 2
        x1c, y1c = max(0.0, x1), max(0.0, y1)
        x2c, y2c = min(1.0, x2), min(1.0, y2)

        if x2c <= x1c or y2c <= y1c:
            continue

        new_boxes.append((cls, (x1c + x2c) / 2, (y1c + y2c) / 2, x2c - x1c, y2c - y1c))

    return out_img, new_boxes
```

- Scale/zoom an image about its center and rescale YOLO-format boxes to match.
- scale > 1 -> zoom IN (simulates a closer vehicle): resize up, then center-crop.
- scale < 1 -> zoom OUT (simulates a farther vehicle): resize down, then center-pad.
- scale == 1 -> no change.

2. Chosen Intensity Values With Justification

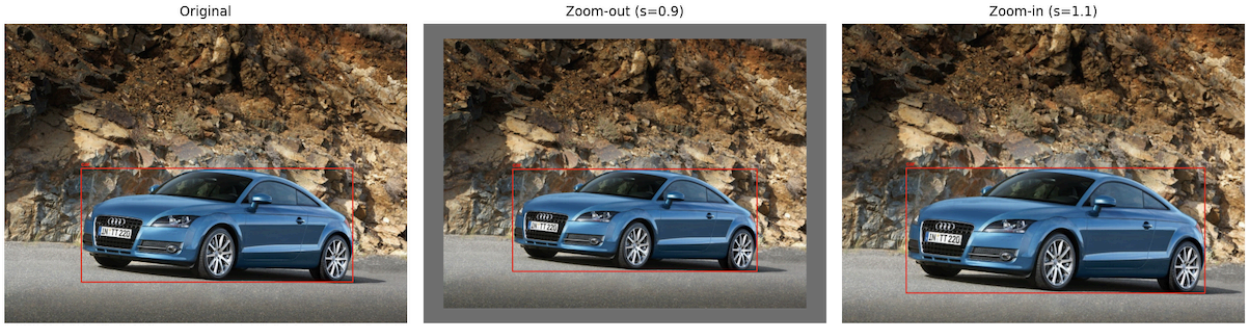
Intensity	Scale Range	Justification
Low	[0.90, 1.10]	±10% size change mimics small natural distance variation between similarly positioned vehicles (e.g., adjacent lanes). A gentle augmentation that shouldn't destabilise training.
Medium	[0.75, 1.25]	±25% size change approximates the apparent size difference between a vehicle one car-length closer/farther, a common real-world range in traffic camera footage.
High	[0.60, 1.40]	±40% size change deliberately aggressive, simulating near-field vs. far-field vehicles (e.g., a truck close to the

		camera vs. one near the vanishing point of the road). Tests whether YOLO's multi-scale detection heads degrade under a wide, unrealistic at the extreme size range.

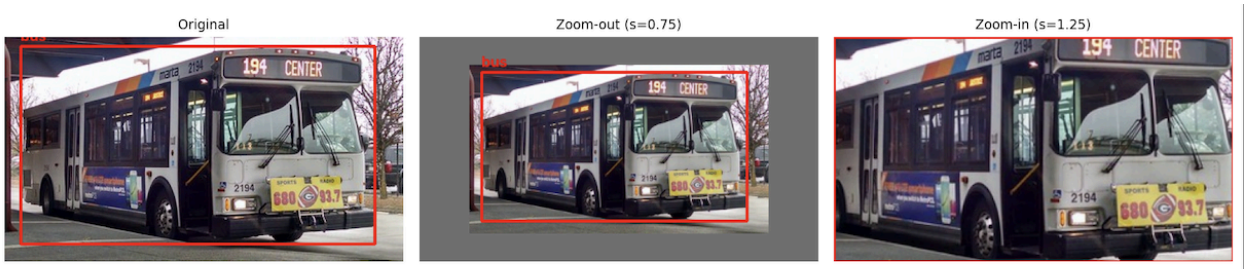
3. Before/After Visual Samples

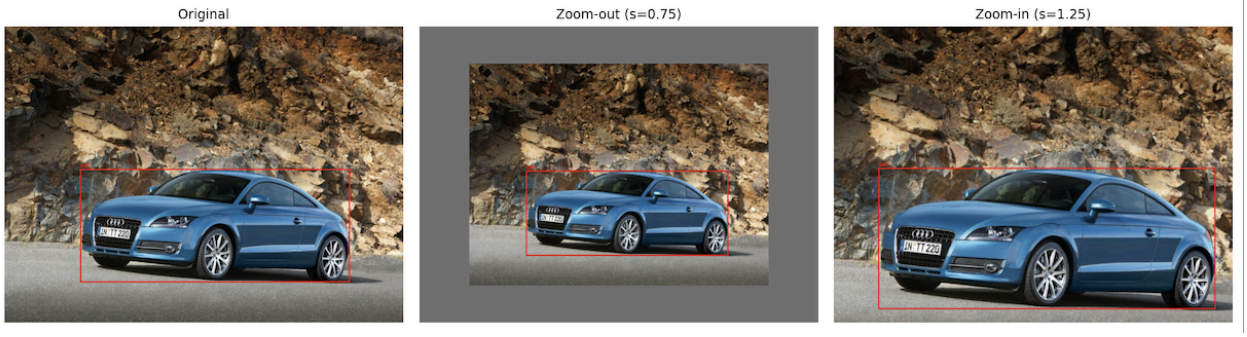
A. Low Intensity



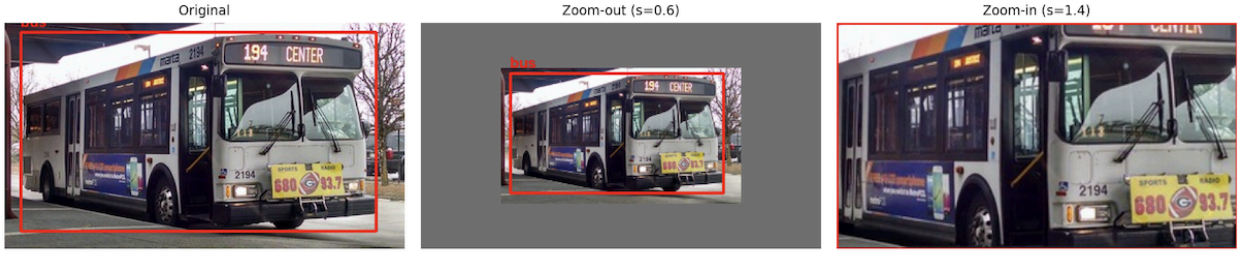


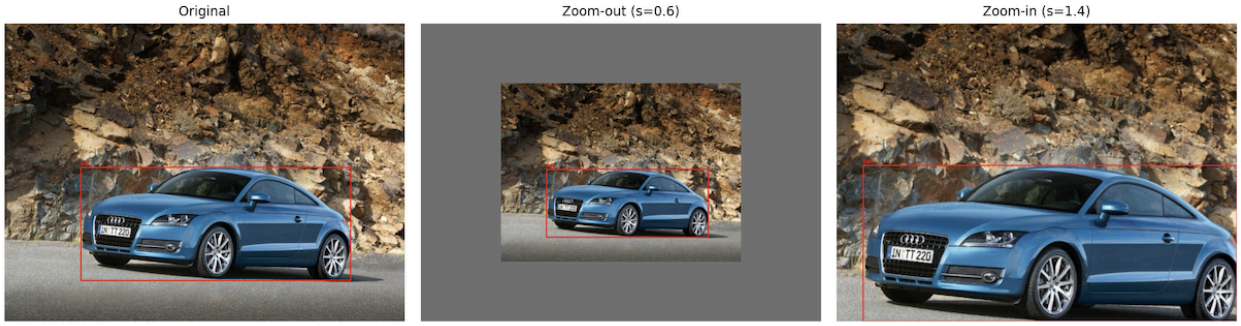
B.Medium Intensity





C.High Intensity





ISSUES/CHALLENGES:

- We had to make sure the bounding box coordinates were recalculated correctly for both zoom-in and zoom-out, since a small error here would silently mislabel every augmented image.
- We needed to decide what to do with boxes that partially or fully leave the frame after zooming in, settling on trimming partial boxes and dropping ones that fall completely outside.
- Choosing the intensity ranges required some judgment since there's no fixed rule for what counts as "Low/Medium/High"; settled on values based on realistic near/far vehicle size differences.

PLAN FOR NEXT WEEK:

- Apply the scaling/zoom augmentation across the full training set for each intensity level (Low, Medium, High), generating three complete augmented datasets.
- Keep dataset split, image size, epochs, batch size, and hardware identical to the Week 1 baseline runs.
- Train YOLOv8n, YOLOv11n, and YOLOv26n separately on each augmented dataset version.
- Save trained weights for every intensity level across all 3 YOLO versions.
- Document everything in the Week 3 report.

HOURS SPENT (PER MEMBER):

Members	Hours Spent	Contribution
Simran Koshta (B230707CS)	4	Studied scaling/zoom theory and derived the bounding-box transformation math for center-based scaling, Report.
Ardhra Alaina Winston (B230198CS)	4	Implemented the scaling/zoom augmentation function in Python/OpenCV, including box clipping/trimming logic, Report.
Maithili Kolasani (B230385CS)	3	Designed and justified the Low/Medium/High intensity scale ranges, Report.
Gunna Mohana Prathyusha (B230324CS)	3	Generated before/after sample grids and visually verified bounding boxes across all three intensities, Report.
Sreeneha Bhavanasi (B230092CS)	2.5	Reviewed outputs, compiled results, and prepared the weekly report.